GLASS Imaging

Neural ISPs (Image Signal Processor)
IEEE SD - 2026

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The GLASS approach

GLASS AI is a proprietary Neural ISP software developed on the principles of computational photography. Tailored specifically to each camera model, our neural networks learn to reverse lens aberrations, sensor effects and noise, delivering remarkable Image Quality efficiently on the edge.

We work with OEMs to train and seamlessly deploy our GLASS AI modes on any smartphone / any camera hardware.

GLASS AI

Global Presence

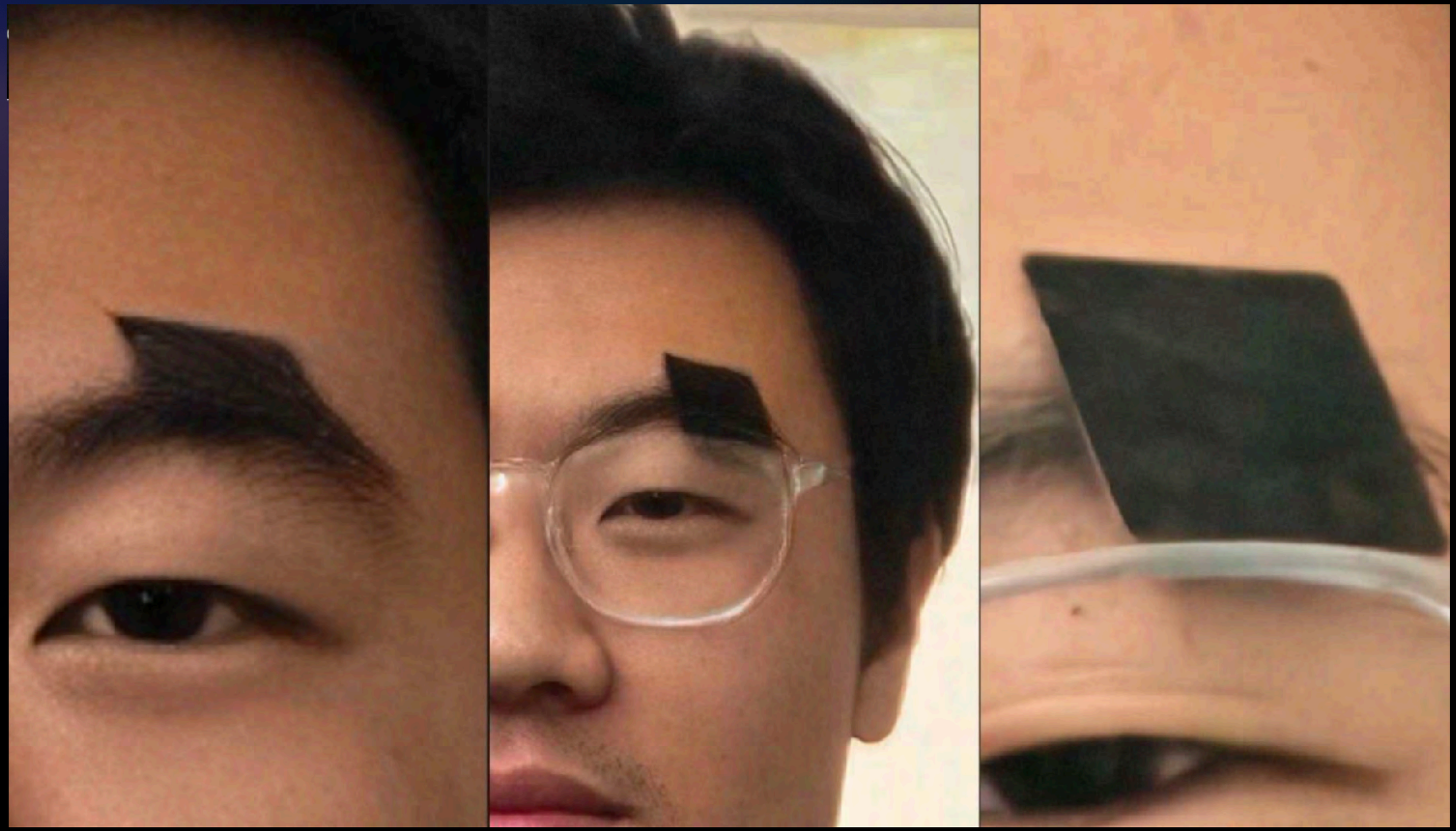
Partners: Google QUALCOMM

Qualcomm Partnership: Ready-to-Go for Smartphones

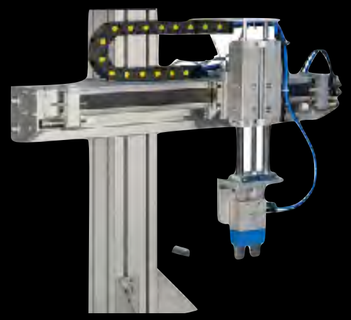
Supported Chipsets:

- SM8650 SM8635
- SM8750 SM8850 (Latest Snapdragon 8 Elite Gen 5)

The Glass Library is optimized to leverage the Qualcomm HTP (Neural Engine)



CAMERAS ARE EVERYWHERE



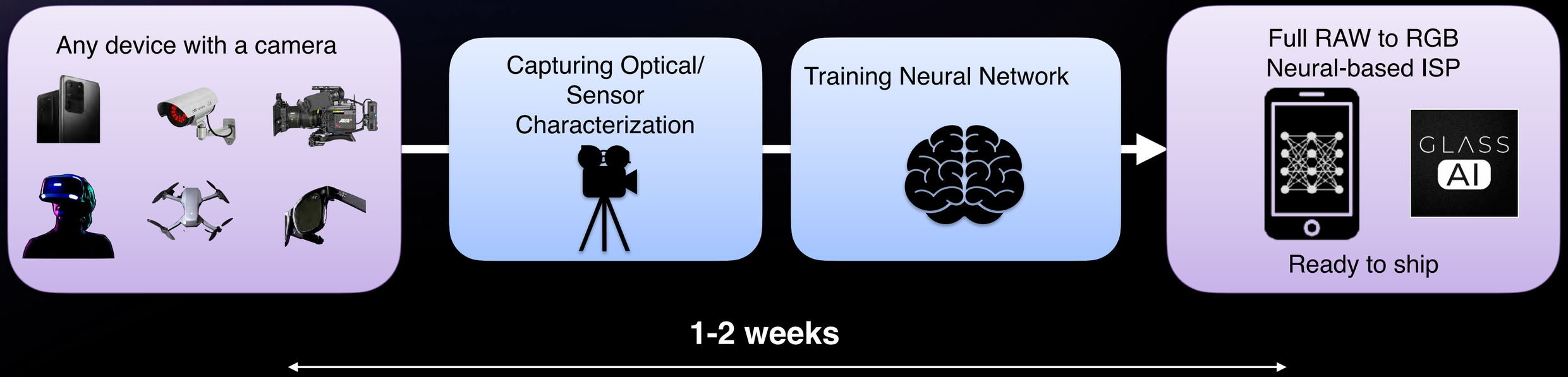
How to rapidly bring optimum image quality to any camera?

Any device with a camera

?

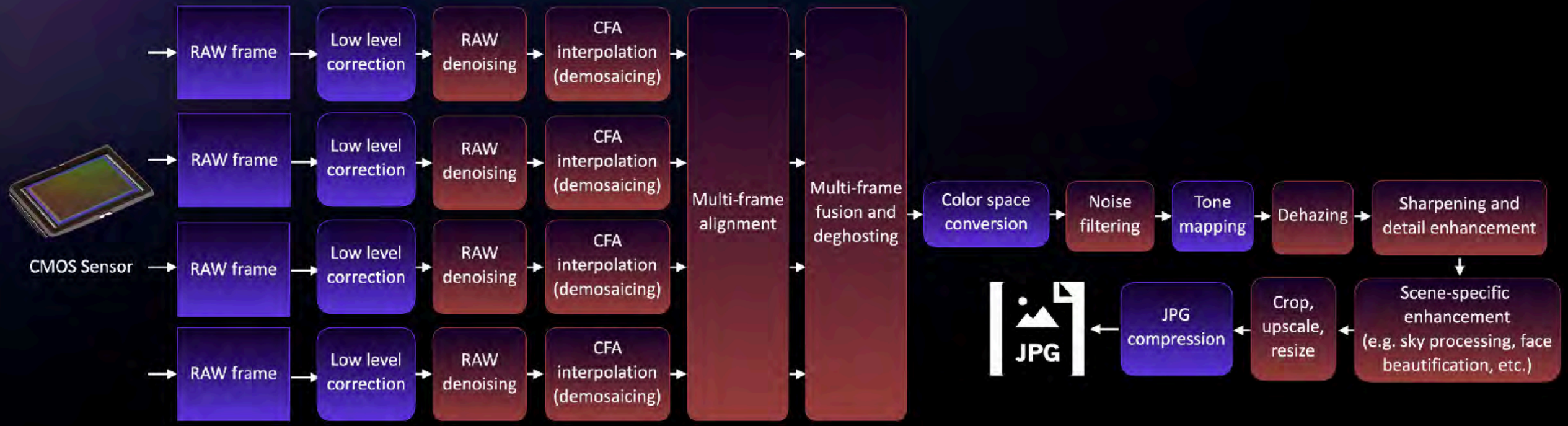
Custom edge neural network deployed on device

From months of per-device tuning to a learning system that adapts!



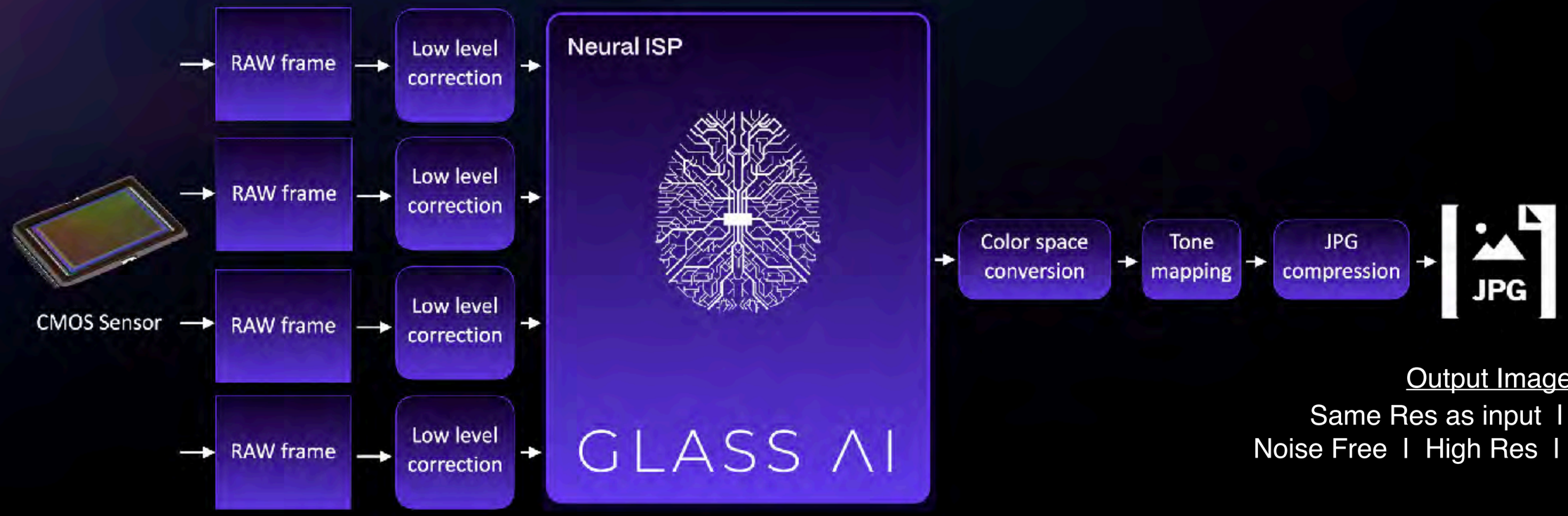
For comparison, competitor company employs thousands of engineers, dedicating multiple months each year to similar development efforts.

A Typical ISP pipeline is complex and difficult to develop



All RED blocks are replaced by Glass Single Neural Network

Imaging pipeline using Glass AI is simple and effective



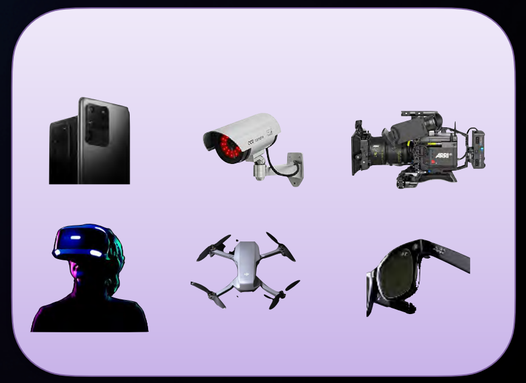
Output Image :

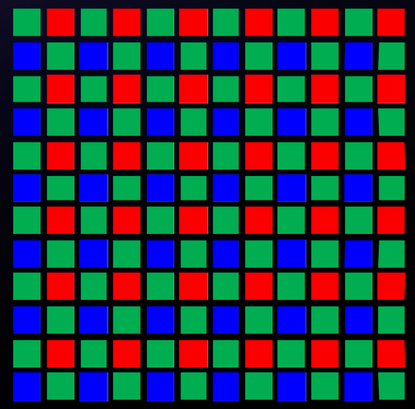
Same Res as input I HDR I
Noise Free I High Res I HD Colors

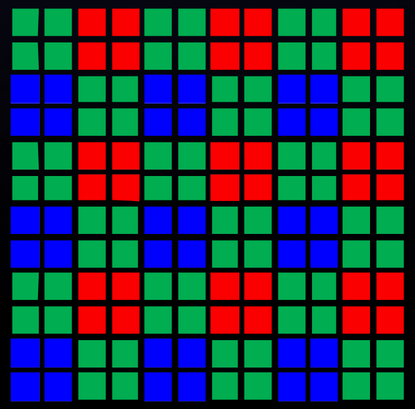
All RED blocks were replaced by Glass Single Neural Network

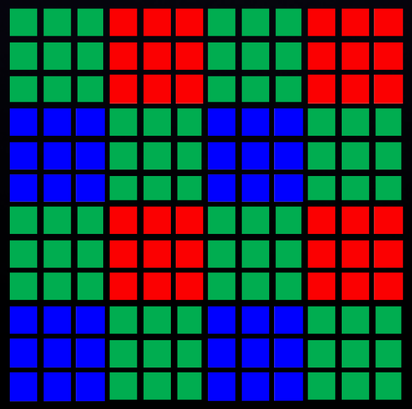
Supports

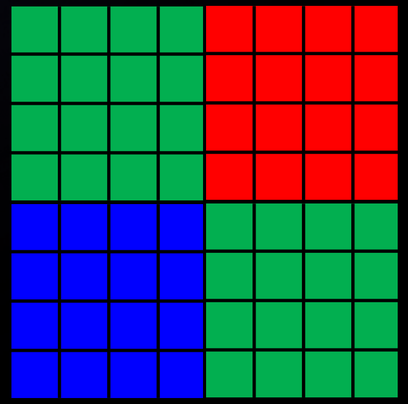
- All pixel sizes
- Any CFA type (RGB, RGBW, …)
- Any CFA pattern (bayer, quad, hex…)
- Any type of lens modules (wide, tele, super-tele (super-high zoom)…)

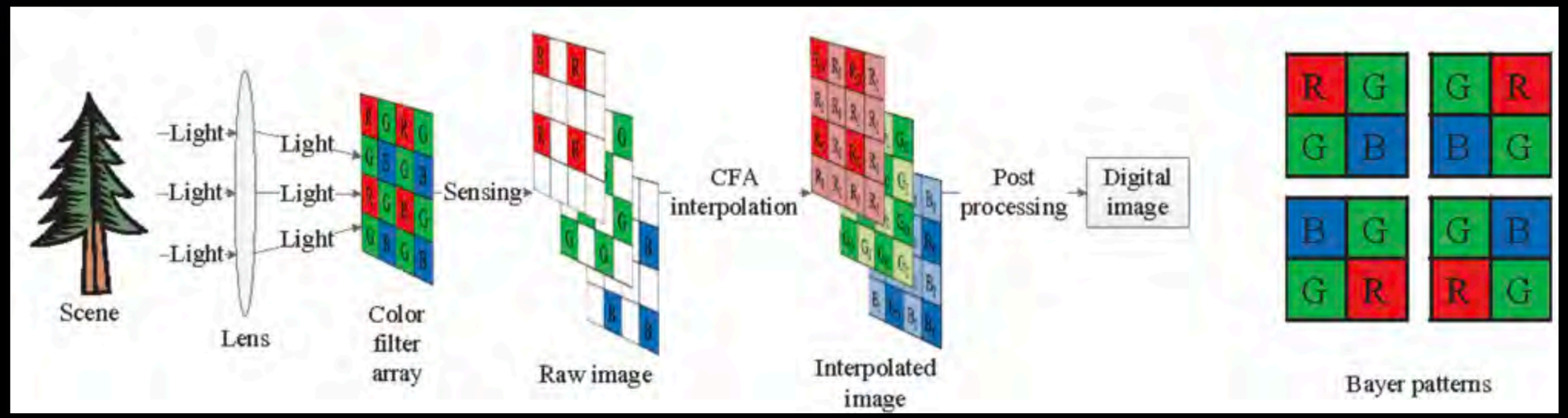












Neural ISP Block

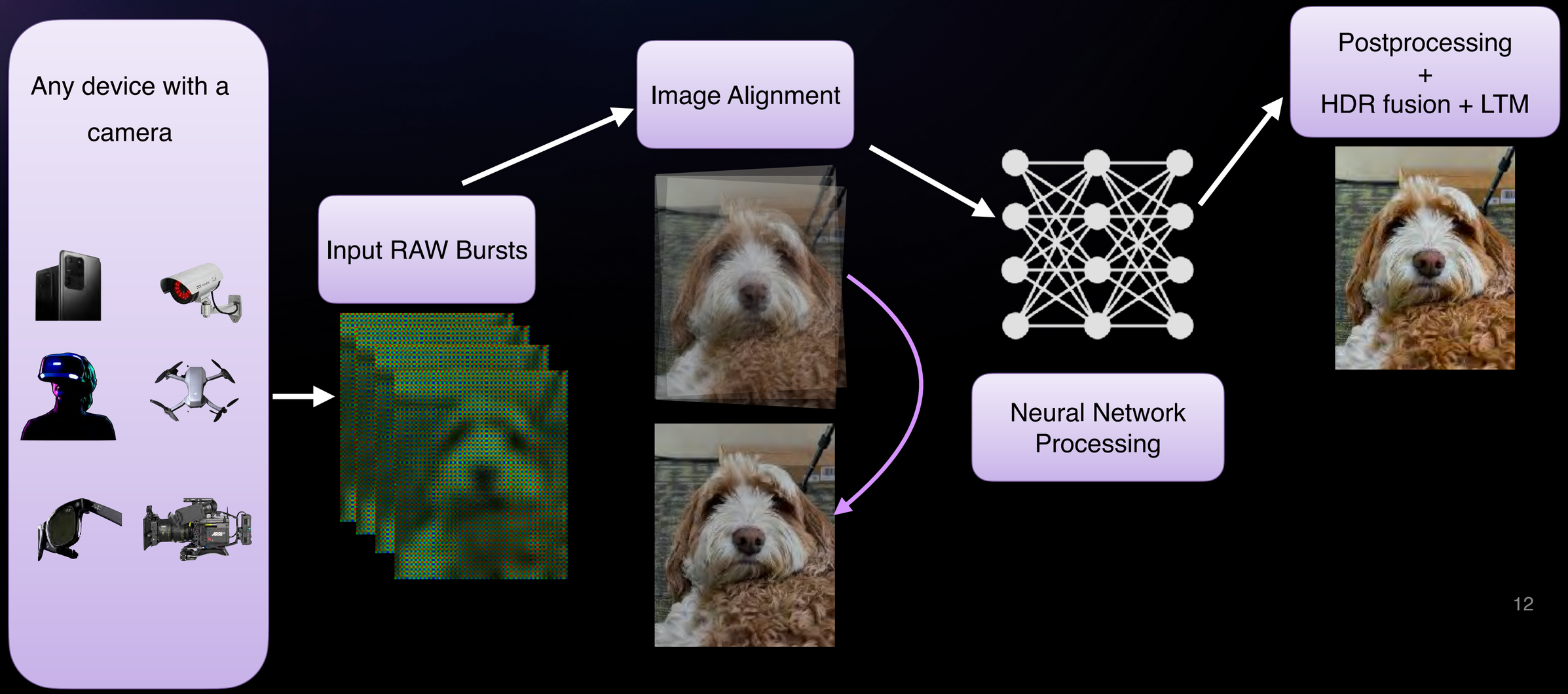
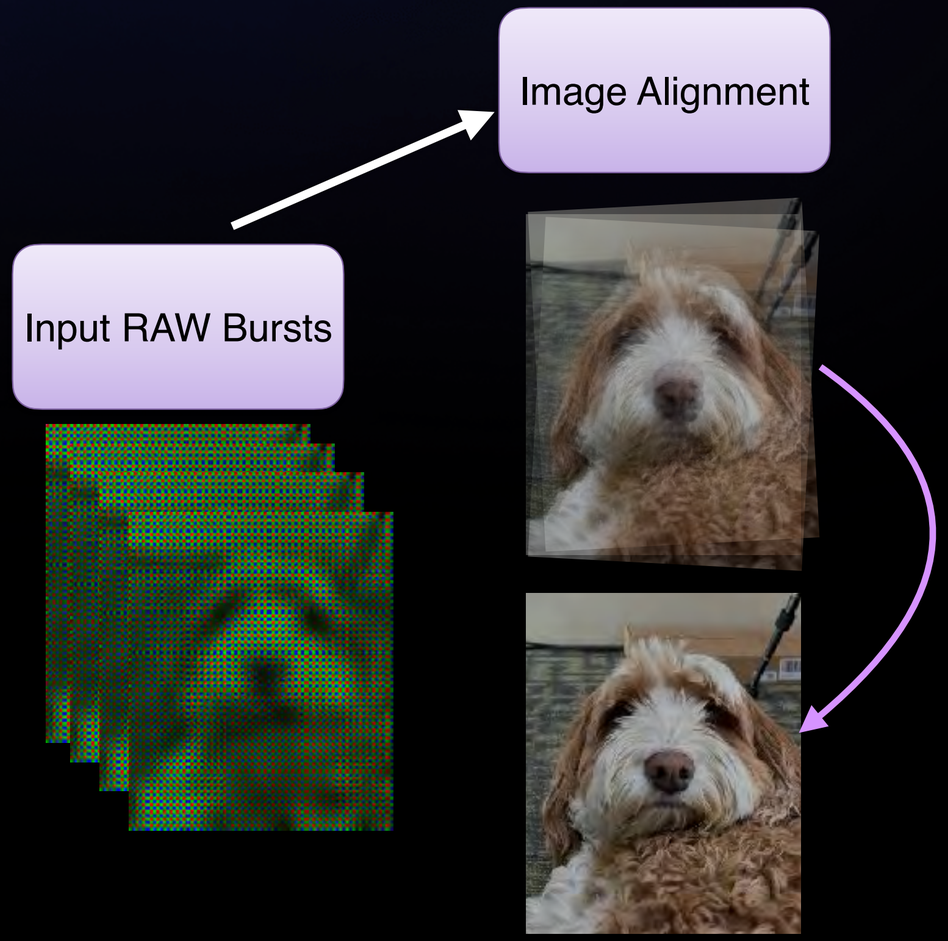


Image Alignment



Evaluation protocols : Not as straightforward as it may seem!

Some existing metrics in academic research

1. PSNR (Peak signal to noise ratio)

$$\mathrm{PSNR} = 10 \cdot \log_{10}\left(\frac{MAX_I^2}{\mathrm{MSE}}\right)$$

2. SSIM (Structural Similarity Index Measure)

$$\mathrm{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

3. LPIPS (Learned Perceptual Image Patch Similarity)

$$LPIPS(I, \hat{I}) = \sum_{l \in L} w_l \cdot \frac{1}{H_l W_l} \sum_{h,w} \left\| f_I^l(h,w) - f_{\hat{I}}^l(h,w) \right\|_2^2$$

4. TOPIQ (Transformer-based Opinion-aware Image Quality)

$$Score = F_{TOPIQ}(I;\theta)$$

User's Preference

1. Prefer fine-grained noise — perceived as sharpness contradictory to smoothness from PSNR

2. Desire for an aesthetically pleasing, branded look — contradictory to perfect structural similarity.

3. Desire realistic detail but no hallucinated texture, sometimes may even prefer smoothening of overly sharp realistic details.

4. Don't want a "generically good" photo; they want a photo with a specific **style** or **look**. Ex: iPhone look or Pixel look or Huawei look

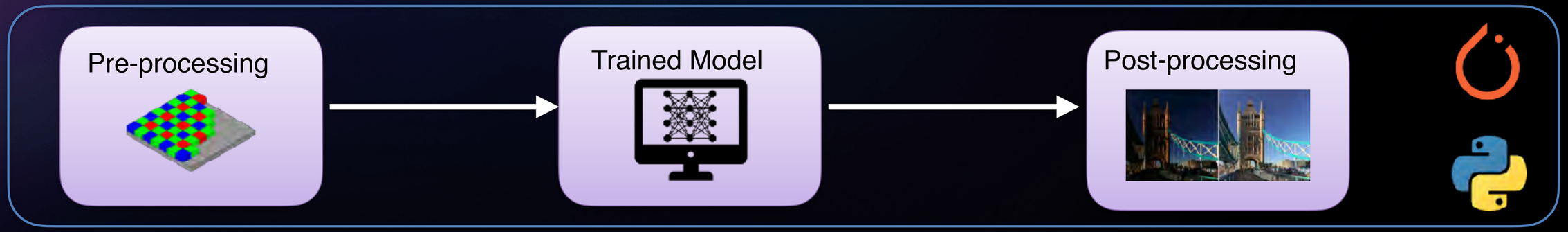
The Optimization Challenge: SOTA Models vs. The Edge

Many state-of-the-art (SOTA) models, while achieving high metric scores, are too large and computationally expensive for practical, real-time deployment on Edge devices like android smartphones.

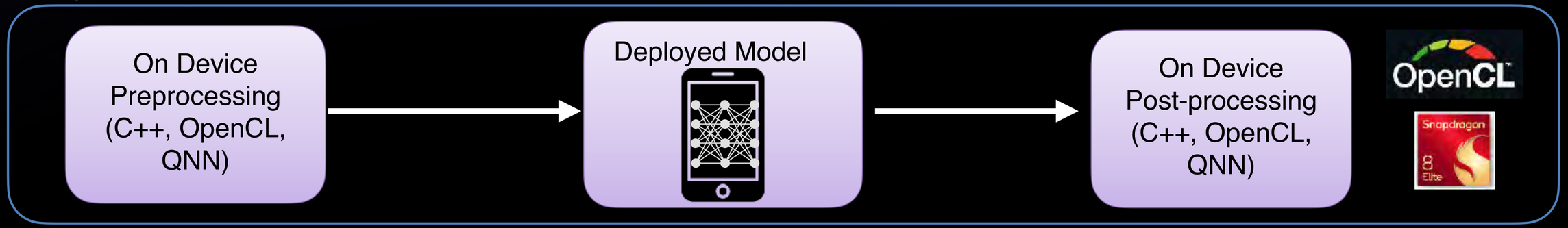
Model	Size	Rough Runtime (512x512 on RTX 4090)
BSRT	~20M/ 5M (B/S)	~100-150ms
Restormer	25.31M	~90-100 ms
Uformer	~50M/ 20M/5M (B/M/S)	~125-140ms

How do we deploy onto your phones?

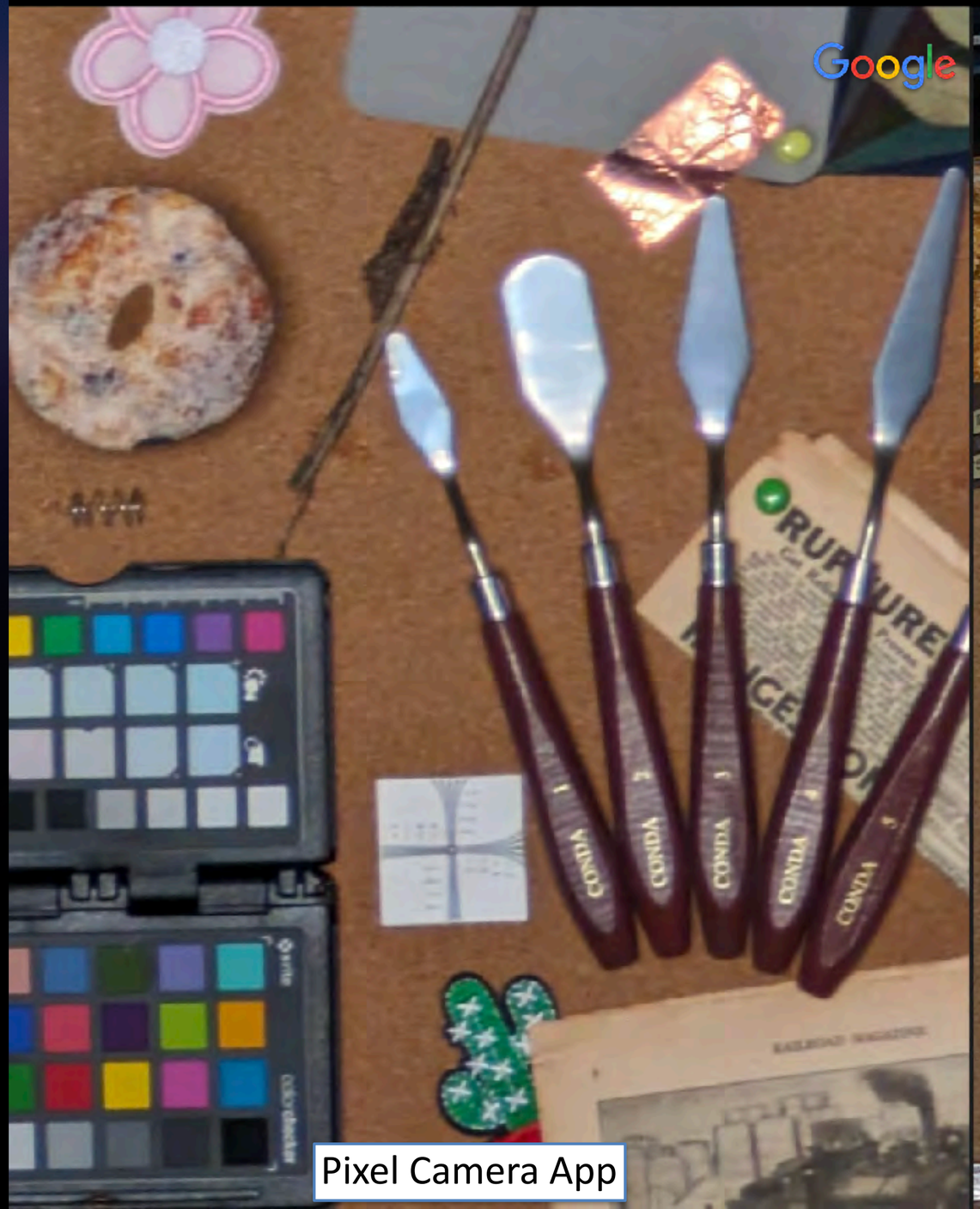
Development:

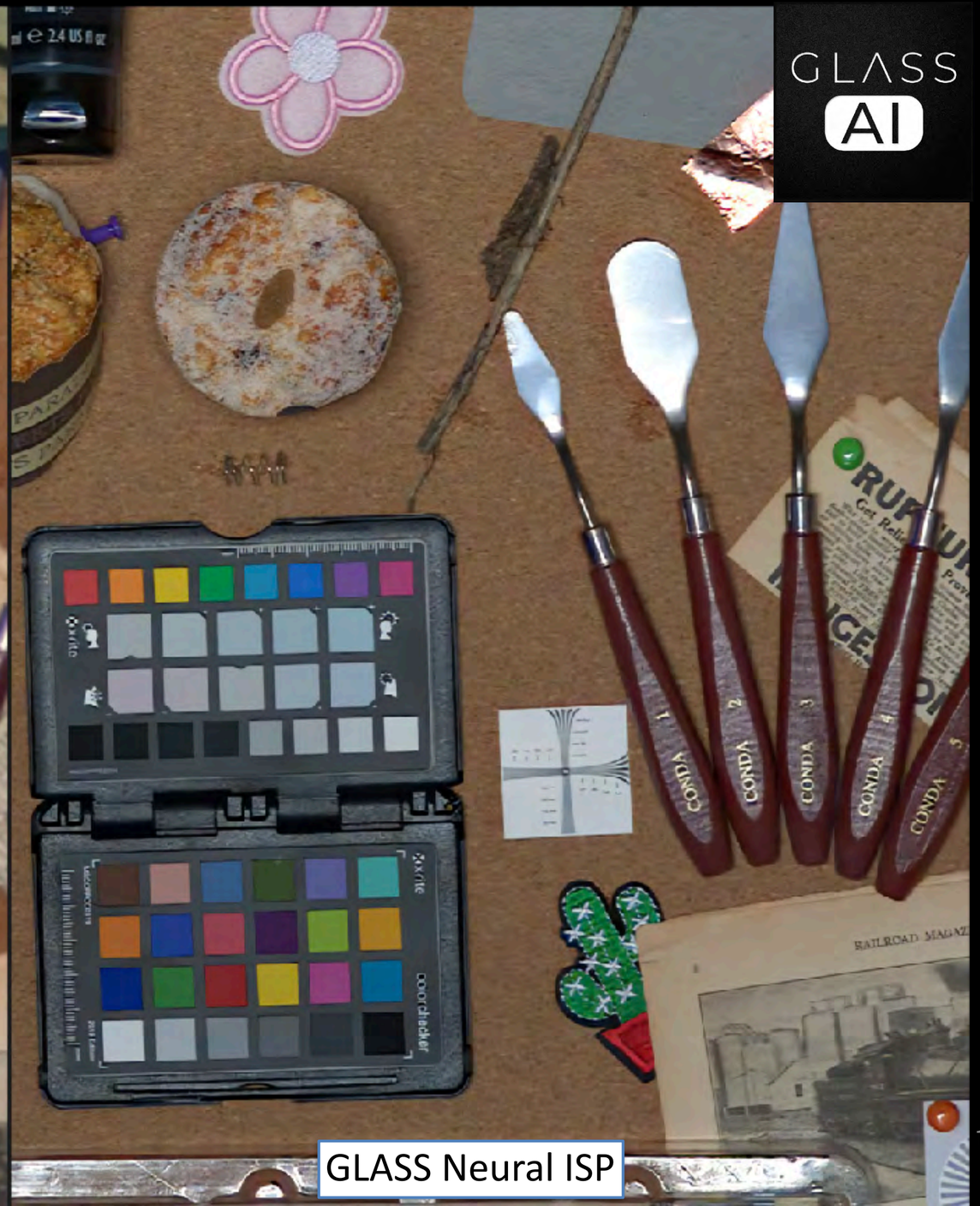


Deployment:



* Operations must be translated to on device **C++**
* Translating **PyTorch** to **OpenCL** + **QNN** isn't trivial


Google
Pixel Camera App


GLASS
AI
GLASS Neural ISP

200MP Hex mode - Android Flagship - Default

200MP Hex mode - Flagship Android Phone
+ GLASS Neural ISP

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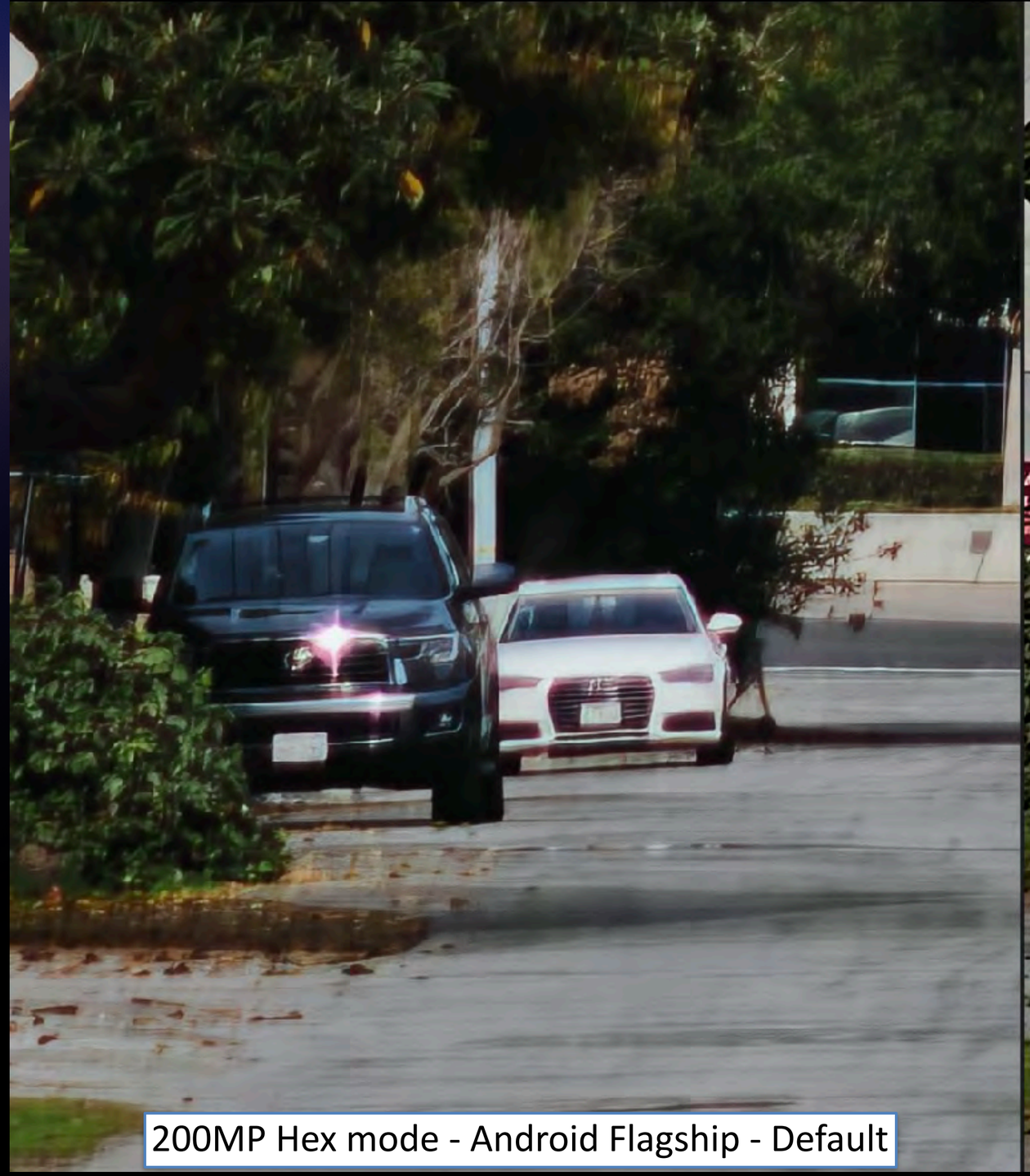
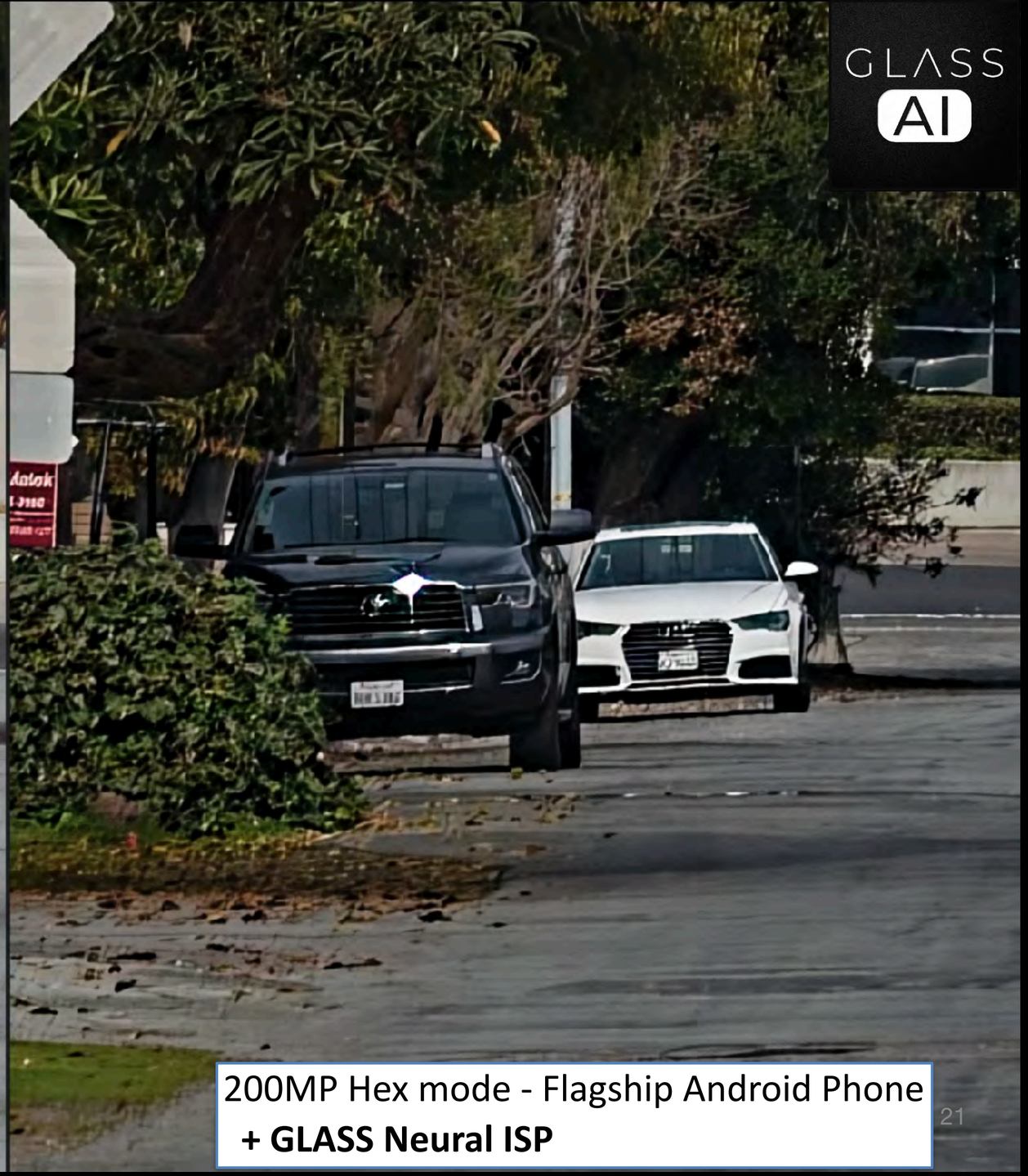
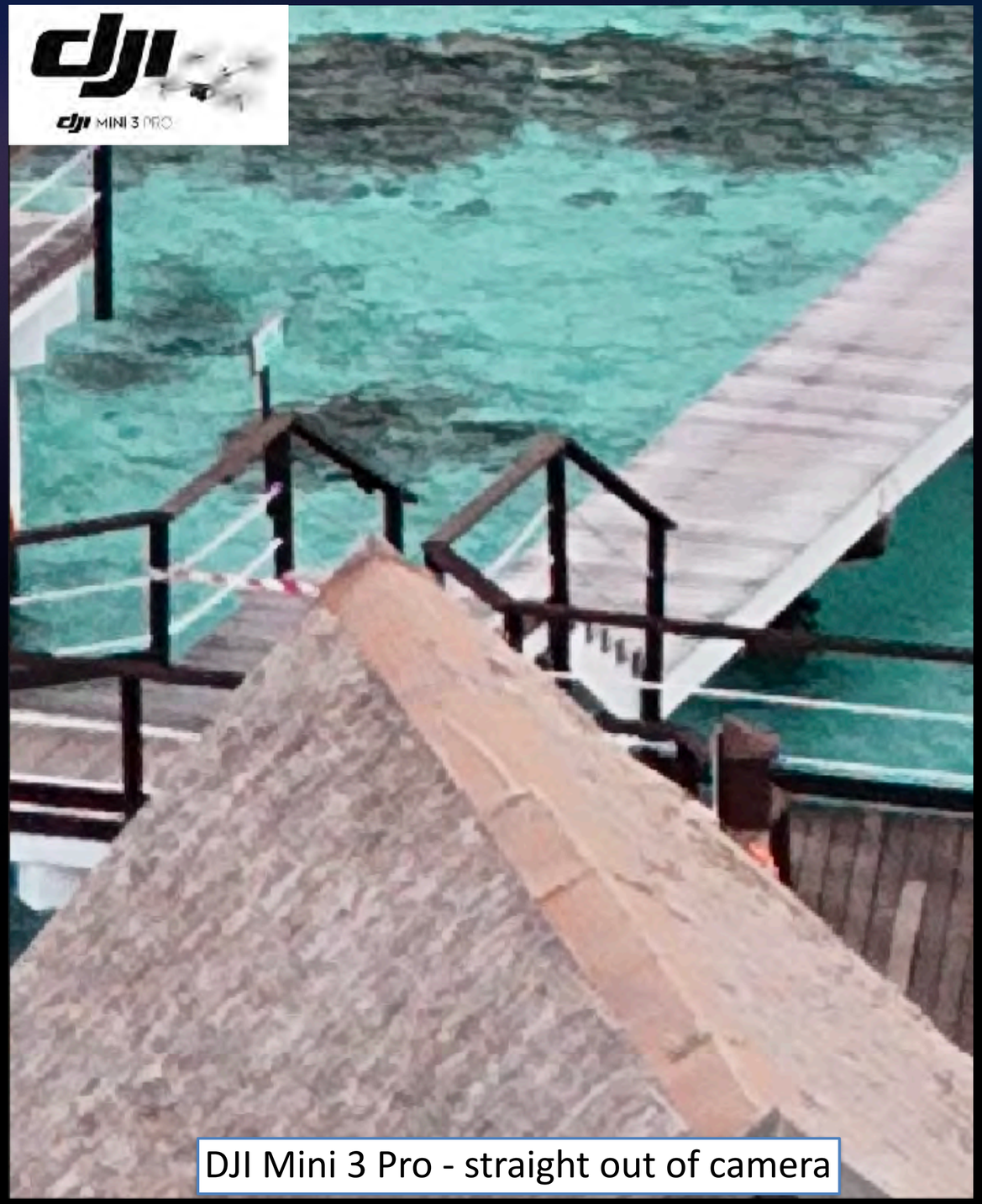


200MP Hex mode - Android Flagship - Default

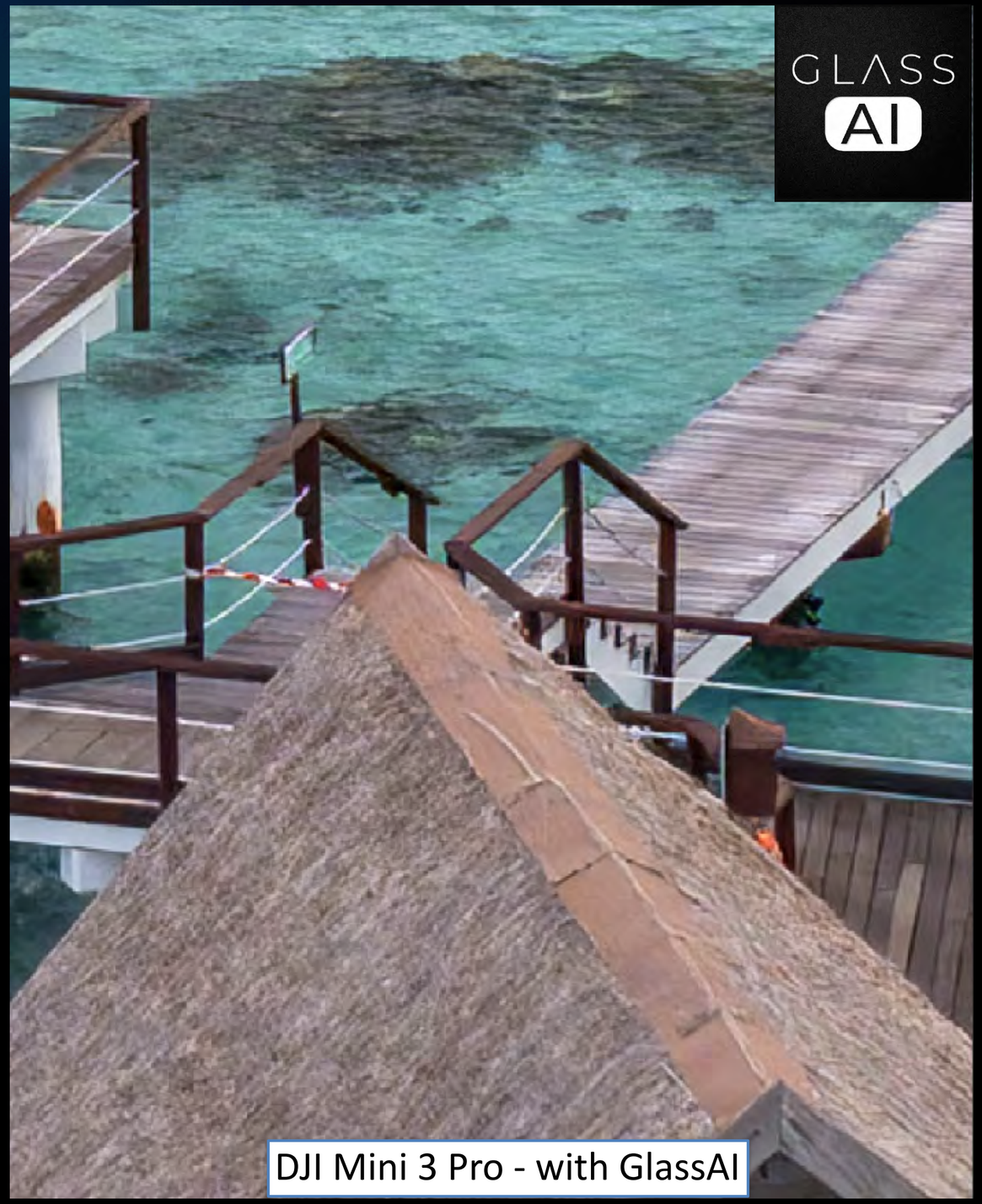


200MP Hex mode - Flagship Android Phone
+ GLASS Neural ISP

GLASS AI



DJI Mini 3 Pro - straight out of camera



DJI Mini 3 Pro - with GlassAI

DXO Mark - Motorola Edge 40 Pro with Glass AI (3rd party objective evaluation)

Ranking	Device	Tele score
1st	Huawei P60 Pro	118
4th	Vivo X90 Pro Plus	115
9th	Google Pixel 8 Pro	112
14th	**Motorola Edge 40 Pro (Glass)**	**108**
15th	Vivo X90 Pro	107
20th	Apple iPhone 15 Pro Max	104
26th	Apple iPhone 15 Pro	103
87th	Motorola Edge 40 Pro	58

Glass AI

108

Total Tele Score

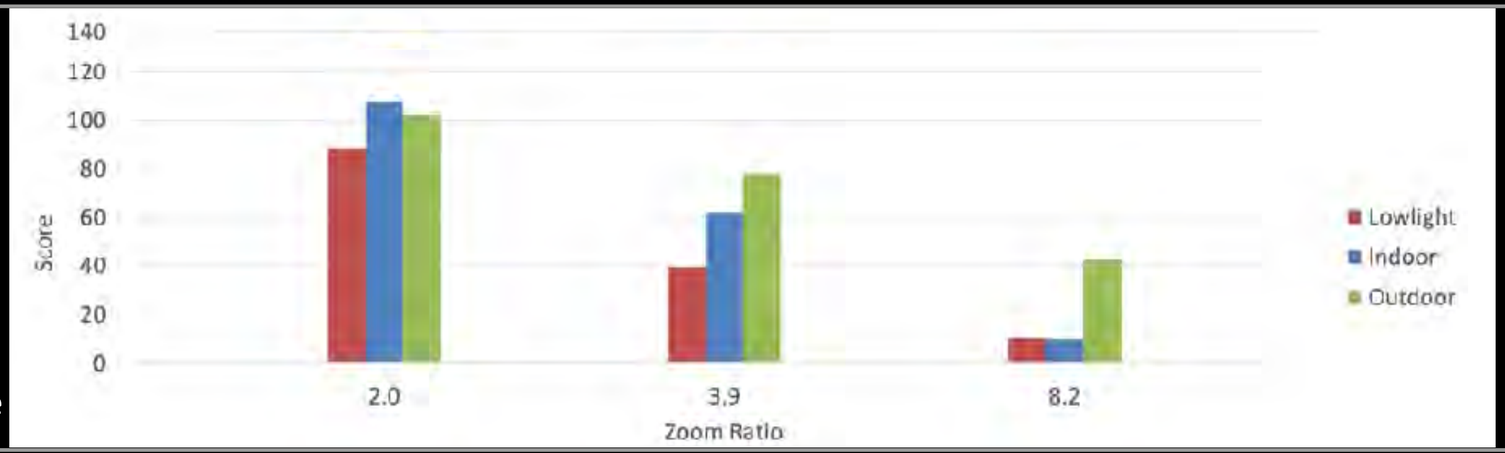
With Glass AI

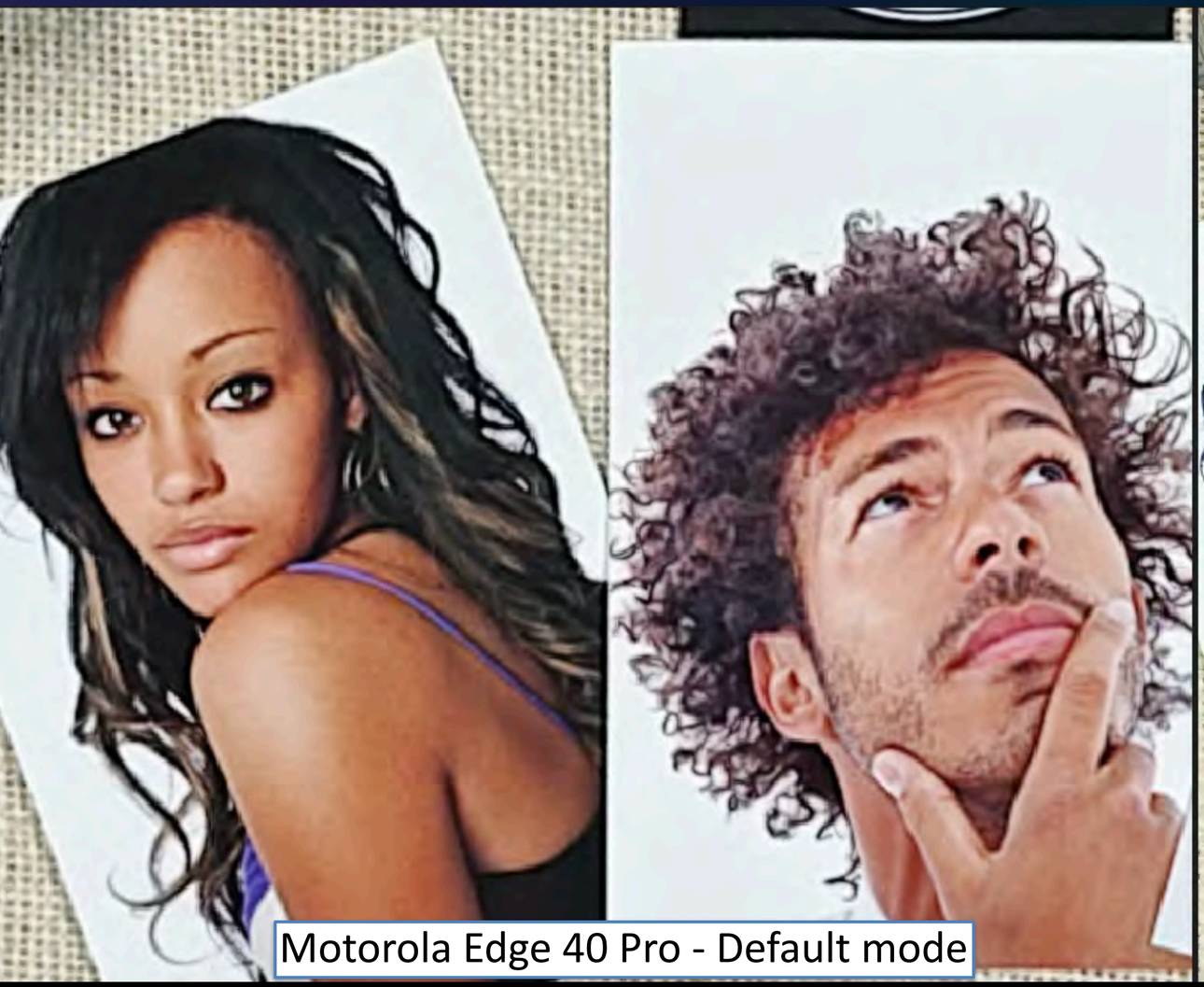
Default App

58

Total Tele Score

Original Motorola Score



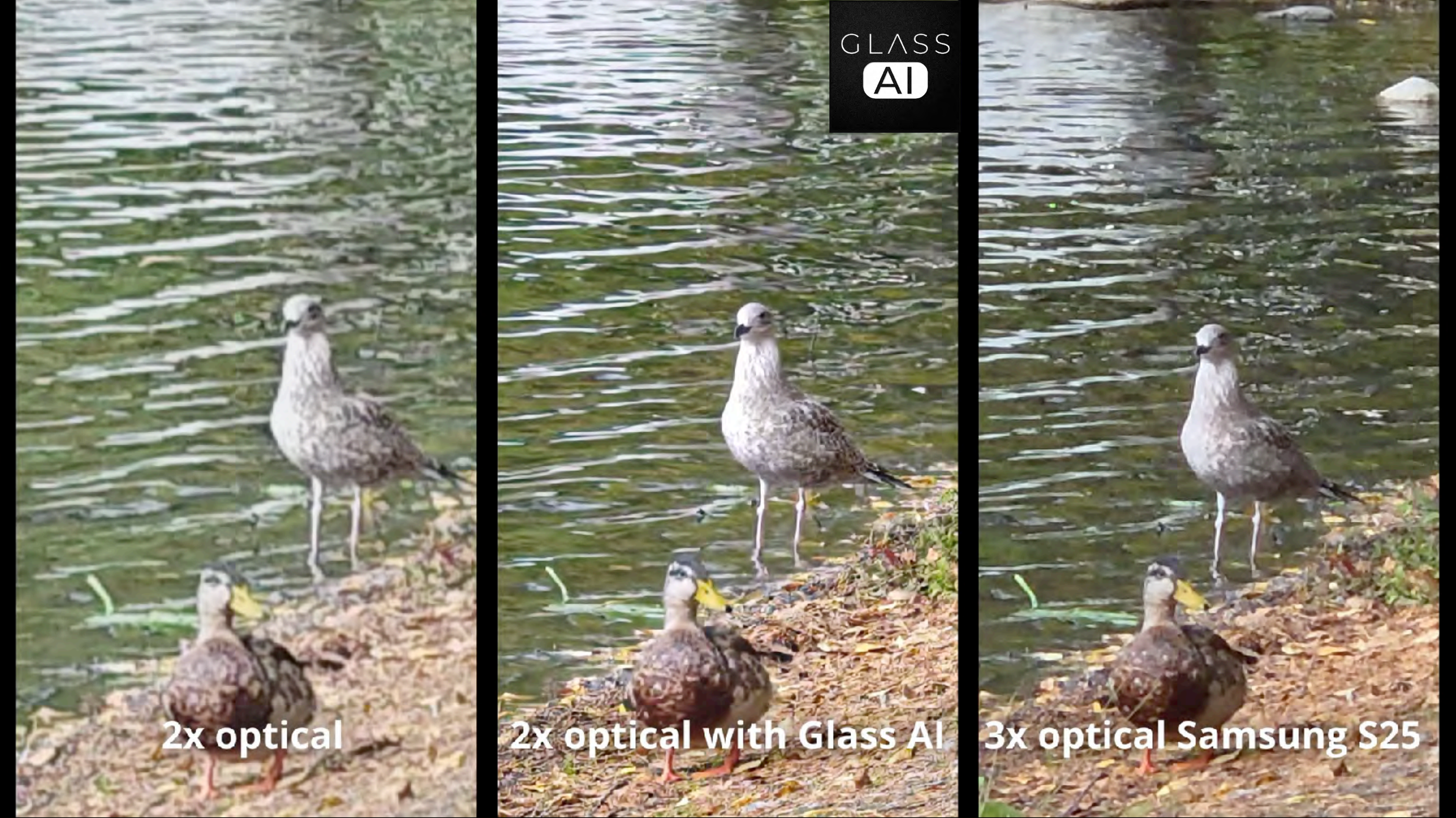


Motorola Edge 40 Pro - Default mode

Motorola Edge 40 Pro **+ GLASS Neural ISP**

Any device with a camera

This is just the beginning of what we're building at GlassAI with still photography — next up is video, where we're already seeing equally strong gains!


GLASS
AI
2x optical
2x optical with Glass AI
3x optical Samsung S25

GLASS
AI
2x optical
2x optical with Glass AI
3x optical Samsung S25

The New Paradigm in Computational Imaging

- **World's best camera ISP software**

  Fixes lens and sensor flaws on any camera, from phones to drones. Crisp, clear images everywhere

- **End-to-End Learned Neural ISP**

  Replaces legacy camera SW pipelines (Demosaicing, denoising, deblurring, color correction)

- **Dramatically Accelerated camera R&D**

  Ship new imaging features in weeks instead of months—no huge tuning teams required

- **Next-Gen Software-Driven Optics**

  Enabling lighter, cheaper, lower-power cameras & radical lens designs

Thank you for your time!
Feel free to reach out at shivansh@glass-imaging.com to learn more about our upcoming work and exciting developments.